NANDA KISHORE K

Software Engineer | AI Engineer
knandakishore6@gmail.com | +91 8281588640 | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience building AI-powered backend systems, Voice AI agents, and scalable microservices. Skilled in LLMs, RAG, conversational AI, and real-time voice systems using OpenAI, Gemini, and LiveKit. Proficient in Python, FastAPI, PostgreSQL, Docker, and event-driven architectures.

TECHNICAL SKILLS

- **Voice & Conversational AI:** Conversational AI Agents, RAG Pipelines, Real-Time WebRTC, LiveKit, STT/TTS Pipelines, Function Calling, Prompt Engineering, Knowledge Base Systems
- **AI Frameworks & Agentic Systems:** LangChain, LangGraph, Agentic AI, Tool Calling, Intent Classification, Structured Outputs, Context Engineering
- **AI Ecosystem & Platforms:** Cloud LLM APIs (OpenAI, Gemini), Voice AI Cloud Platforms (Deepgram, Cartesia), Local LLM Runtimes & Models (Ollama, Gemma, Qwen, Whisper)
- **Backend & Systems:** FastAPI, Node.js, Laravel, Microservices, RESTful APIs, WebSockets, Event-Driven Architecture, Celery, Celery Beat, Redis Queue, JWT/OAuth, Razorpay Integration
- **Databases & DevOps:** PostgreSQL, MySQL, SQL Optimization, Data Modeling, Indexing, Docker, Nginx, PM2, Linux (Ubuntu), Git, SSH, Systemd
- **Languages & Frameworks:** Python, JavaScript, PHP, Dart, C++, React.js, Flutter (Web & Mobile), Electron, HTML5/CSS3, Bootstrap

PROFESSIONAL EXPERIENCE

Cognidax | *Software Engineer*
Oct 2025 – Present

- Architected high-concurrency backend microservices using FastAPI and Node.js for a production **Cloud Telephony AI Platform**, handling low-latency audio streams and full-duplex WebRTC/WebSocket communications.
- Built end-to-end Retrieval-Augmented Generation (RAG) pipelines and LLM Function Calling workflows using cloud models (OpenAI, Gemini) and self-hosted models (Ollama, Gemma, Qwen) for real-time automated phone agent decisions.
- Implemented custom Speech-to-Text (STT) and Text-to-Speech (TTS) streaming channels (Whisper, Deepgram, Cartesia) and integrated payment gateways (Razorpay) with event-driven webhooks for automated **over-the-phone subscription checkout and recurring billing verification**.
- Orchestrated asynchronous telephony event queues and task processing via Celery, Celery Beat, and Redis Queue while optimizing state storage using PostgreSQL query tuning and indexing.
- Containerized and deployed telephony services using Docker, Nginx, and PM2 on Linux (Ubuntu) environments with Systemd process monitoring.

Reino Technologies | *Software Engineer*
May 2024 – Sep 2025

- Developed secure RESTful APIs and server-side logic using Laravel for enterprise-grade government applications and real-time AI communication platforms via LiveKit/WebRTC.
- Executed SQL query performance tuning, database indexing, and schema updates on high-volume relational databases to minimize latency under load.
- Integrated cross-platform client interfaces using Flutter, establishing seamless API contracts between backend infrastructure and mobile/web surfaces.

Amrita Vishwa Vidyapeetham | *Junior Software Developer*
Apr 2022 – Apr 2024

- Engineered web application backend services and structured REST APIs using Laravel and PHP for internal management systems and university portals.
- Maintained relational databases (MySQL), authoring complex database queries, schema migrations, and transactional procedures.
- Assisted in end-to-end application maintenance, deployment troubleshooting, and server environment optimization.

TECHNICAL PROJECTS

- **YouTube AI Creator Assistant:** Developing an AI-powered YouTube creator platform for comment management, intent-based reply automation, and human review. Building conversational AI workflows using Python, FastAPI, LangChain, and LangGraph, with PostgreSQL persistence, JWT authentication, Google OAuth, multi-channel support, conversation history, context management, and long-term memory.
- **AI Portfolio & Autonomous Interview Simulator:** Built a web-based portfolio with an autonomous voice/text assistant using a custom RAG engine. Features an automated preliminary recruiter interview simulator.
- **Local LLM & Privacy-Focused AI Playground:** Developed a privacy-focused environment deploying local LLMs (Ollama, Gemma, Qwen, Whisper) integrated into Cursor/Claude workflows for offline code analysis.
- **Embedded Edge Voice AI Hardware Client (Raspberry Pi):** Engineered a lightweight, edge-deployed voice processing system on Raspberry Pi hardware. Configured low-level audio subsystems (ALSA/PulseAudio), hardware I/O peripherals, and DSP audio buffering to execute real-time local STT/TTS inference with minimal latency.
- **Electron Desktop Kiosk Application:** Developed cross-platform desktop kiosk apps using Electron with deep local API integration, secure application state management, and automated background sync.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering
Govt.College of Engineering, Cherthala | 2017 – 2021

CERTIFICATIONS

• Database Management Systems – NPTEL
• Front-End Development – Meta